Course Code : ECT 452-2 **UVXV/MW – 24 / 1666**

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

BROADBAND COMMUNICATION

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory.
(2) All questions carry marks as indicated against them.
(3) Due credit will be given to neatness and adequate dimensions.
(4) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (a) Deduce with the help of various H channels, how the B-ISDN satisfies need of every customer requirements ? 5(CO1)

(b) Describe SONET with the help of its general configuration. 5(CO1)

2. (a) Elaborate in short the evolution of Satelite Communication. Also explain Clark's statement about Possibility of Satellite communication. 5(CO2)

(b) What are various orbits in which Satellite revolves ? Also describe steps involved in Satellite launching. 5(CO2)

3. (a) Prove that height of satellite in circular orbit is approximately 3600 Km. 5(CO3)

(b) A scientific satellite is moving on an Earth orbit with perigee altitude of 400 km and apogee altitude of 15000 km.

(i) Calculate the eccentricity of this orbit.

(ii) Calculate the period of this orbit. 5(CO3)

4. (a) List various types of controls required to maintain the satellite in space and explain TTC and M system in detail. 5(CO3)

(b) Describe the role of antenna in satellite communication. What are the different types of antennas used for satellite communication ? 5(CO3)

5. (a) What is downlink ? Explain Rain Effects on Downlink. 5(CO4)

(b) Discuss typical Phenomena in Satelite Communication related to Solar and its different phases. 5(CO4)

6. (a) A satellite at 40,000 km from point on earth surface transmits power of 2 W from an antenna having gain 17 dB direction of observer.

(i) Find the flux density.

(ii) Find Power received by antenna with effective aperture of 10 m^2.

(iii) Gain of receiving antenna. 5(CO5)

(b) Write a short note on Drafting of satellite link budget in clear air and rainy conditions. 5(CO5)